

BASE ESPERANZA, Antarctica

WMO: 889630

Lat: 63.3989S

Lon: 56.9978W

Elev: 24

StdP: 101.04

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-24.9	-22.7	-30.4	0.2	-23.9	-28.0	0.3	-22.0	33.0	-16.2	30.6	-17.4	11.5	230	1.158

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	8.6	4.3	7.2	3.4	6.1	2.8	4.7	7.5	3.9	6.4	3.2	5.4	9.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.6	4.6	4.6	1.9	4.3	3.8	1.4	4.2	3.3	18.4	7.6	16.9	6.6	15.5	5.6	10.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	22.5	20.3	DB	-26.8	12.0	2.4	1.7	-28.5	13.2	-29.9	14.2	-31.3	15.2	-33.0	16.4	(4)
(5)			WB	-27.0	7.1	2.3	1.4	-28.7	8.1	-30.0	8.9	-31.3	9.7	-33.0	10.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.6	1.5	1.1	-1.8	-6.0	-7.4	-10.3	-11.3	-9.7	-6.9	-3.8	-1.0	1.0	(6)	
	DBStd	6.88	1.92	2.91	4.44	5.84	6.01	6.75	7.06	6.53	6.62	5.39	3.18	2.04	(7)	
	HDD10.0	5321	263	250	365	480	539	608	661	610	506	429	330	280	(8)	
	HDD18.3	8362	522	483	623	730	797	858	919	868	756	687	580	538	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.3	5.7	6.7	7.3	7.6	7.1	7.8	7.8	7.9	8.3	8.1	7.1	6.3	(14)	
(15) Precipitation	PrecAvg	516	47	44	57	42	42	37	36	44	46	43	45	41	(15)	
	PrecMax	1202	140	123	122	110	110	95	90	153	207	184	146	110	(16)	
	PrecMin	181	2	0	3	4	1	3	0	0	0	7	5	4	(17)	
	PrecStd	229	34	29	31	25	27	24	24	38	40	36	32	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.2	9.9	7.0	9.6	5.6	5.5	4.5	6.8	7.5	8.2	8.3	(19)	
		MCWB	5.6	5.3	5.3	3.0	4.5	1.5	1.9	1.2	2.8	3.1	4.1	3.9	(20)	
	2%	DB	7.3	8.0	7.0	4.6	5.1	3.0	2.2	2.0	4.4	5.5	6.1	6.5	(21)	
		MCWB	3.8	3.9	3.3	2.0	2.0	0.3	0.2	0.1	1.4	2.2	2.7	3.0	(22)	
	5%	DB	5.9	6.7	5.1	3.1	2.6	1.2	0.7	0.7	2.9	4.0	4.5	5.2	(23)	
		MCWB	3.0	3.3	2.5	1.2	0.7	-0.2	-0.2	-0.4	0.7	1.3	1.8	2.2	(24)	
	10%	DB	4.8	5.4	3.8	1.9	1.2	-0.2	-0.5	-0.3	1.5	2.6	3.3	4.2	(25)	
		MCWB	2.3	2.6	1.8	0.6	0.1	-1.1	-1.2	-1.2	0.2	0.7	1.1	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.9	5.7	3.9	5.1	2.0	2.2	1.7	3.4	4.0	4.7	4.7	(27)	
		MCDB	8.7	9.7	9.4	6.0	9.1	4.2	4.6	3.3	6.1	6.2	7.5	7.1	(28)	
	2%	WB	4.2	4.3	3.8	2.4	2.3	0.8	0.6	0.5	1.7	2.6	3.2	3.5	(29)	
		MCDB	6.6	7.1	6.2	4.0	4.4	2.3	1.6	1.6	3.6	4.9	5.5	5.9	(30)	
	5%	WB	3.3	3.6	2.8	1.5	1.1	0.0	-0.2	-0.3	1.0	1.6	2.2	2.6	(31)	
		MCDB	5.3	6.1	4.5	2.7	2.4	1.2	0.7	0.6	2.5	3.5	4.0	4.6	(32)	
	10%	WB	2.6	2.9	2.0	0.6	0.1	-1.0	-1.2	-1.2	0.3	0.9	1.4	1.9	(33)	
		MCDB	4.4	5.0	3.5	1.7	1.1	-0.2	-0.3	-0.3	1.5	2.3	2.9	3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.6	5.2	5.6	6.6	6.4	7.0	7.0	6.9	6.0	4.8	4.2	(35)	
		MCDBR	6.3	7.3	6.7	7.3	8.9	8.0	8.7	8.4	7.7	7.2	6.8	6.1	(36)	
	5% WB	MCWBR	3.9	4.5	4.3	5.4	6.8	6.3	7.5	7.4	5.7	5.0	4.5	3.9	(37)	
		MCWBR	5.9	6.7	6.0	7.0	7.7	7.6	8.4	7.8	7.0	6.6	6.1	5.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.293	0.289	0.275	0.247	0.180	0.153	0.168	0.236	0.264	0.285	0.285	0.292	(40)	
		taud	2.419	2.434	2.439	2.344	2.032	1.923	1.989	2.221	2.301	2.320	2.367	2.375	(41)	
	Ebn at Noon		978	931	855	699	456	257	452	683	843	922	981	992	(42)	
		Edh at Noon	104	91	72	49	26	13	25	53	80	101	109	112	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.72	3.99	2.19	0.92	0.26	0.08	0.17	0.72	2.02	3.85	5.66	6.49	(44)	
	RadStd		0.37	0.21	0.17	0.07	0.02	0.00	0.01	0.05	0.14	0.18	0.29	0.42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page